DESIGNED AND CREATED BY PROJECT MOON PI · USER GUIDE

SkyScope

A live aircraft radar styled after an air-traffic-control display, for a Raspberry Pi on a TV or monitor, or any web browser.



SkyScope centred on Phoenix Sky Harbor at 25 miles: Class B airspace, runways with extended centrelines, arrival routes (amber), aircraft coloured by kind, and the information panel.

Website: projectmoonpi.com/skyscope · Help: support@projectmoonpi.com · Source: github.com/projectmoonpi/skyscope · Licence: MIT · Not for navigation.

Contents

1. What SkyScope does
2. Requirements
3. Installing on a Raspberry Pi
4. Installing on a Mac or Windows PC
5. First-time setup
6. Using the radar
7. The settings page
8. The ATC map
9. Filters and presets
10. Alerts and the watchlist
11. Your own receiver and GPS
12. Troubleshooting

3. Quick reference
4. SkyScope Satellite and Mini
15. Credits and data sources

1. What SkyScope does

SkyScope shows the aircraft around a location you choose, live, on a radar scope. Each aircraft is drawn at its position with a symbol shaped like the aircraft (airliner, business jet, light aircraft, helicopter…), a line showing where it will be in a minute, and a label with its callsign, type and altitude. A side panel lists every aircraft in range with its altitude, distance, bearing and speed, and a card shows details and a photo of the nearest or selected aircraft.

Underneath the traffic you can show an air-traffic-control style map: airports and runways with extended centrelines, US Class B, C and D airspace, and the published departure (SID) and arrival (STAR) routes for airports you pick.

Two ways to display it

Display	Best for	How
Pi display	A Raspberry Pi connected to a TV or monitor, running full screen by itself.	`python3 skyscope.py`
Web radar	Any computer, tablet or phone on your network, or a laptop on its own. Sharp on high-resolution screens; extra features such as clicking aircraft and the filter panel.	Open `http://<address>:7700/radar`, or run `python3 skyscope.py --web-only` on a laptop.

Both run at the same time: a Pi can drive the TV while anyone on the network opens the web radar. Everything is configured from a **settings web page** at `http://<address>:7700/radar_config`, so the Pi needs no keyboard once it's set up.

Where the aircraft data comes from

- **Internet (default):** the free adsb.fi service, which pools reports from thousands of volunteer receivers. Needs an internet connection.
- **Your own receiver (optional):** a USB radio receiver on the Pi picks up aircraft signals directly, with no internet needed. See section 11.

> **Not for navigation.** SkyScope is for watching traffic. Positions can be delayed or missing, and map layers are approximations.

2. Requirements

Raspberry Pi setup

Item	Needed	Notes
Raspberry Pi	Pi 5 (recommended) or Pi 4 with 2 GB+ memory	Older models may work but will be slow at long ranges.
Power supply	Official supply for your model	Pi 5: 27 W USB-C. Pi 4: 15 W USB-C.
microSD card	16 GB or larger	Class A1 or better.
Display	Any HDMI TV or monitor	Widescreen (16:9) is ideal; other shapes work too. A micro-HDMI cable for Pi 4/5.
Network	Wi-Fi or Ethernet	Required for internet data, airline logos and photos.
Keyboard and mouse	For installation only	Afterwards everything is done from the settings page.
Software	Raspberry Pi OS with desktop (64-bit)	Includes Python 3. SkyScope also needs `pygame` and `git` (installed below).

Optional extras

Item	What it adds
USB SDR receiver (RTL-SDR) and a 1090 MHz antenna	Receive aircraft yourself, with no internet. A dongle with a built-in 1090 MHz filter and amplifier (sold for ADS-B) works best. Uses the free `readsb` software.
USB GPS receiver	Centres the radar on the Pi's own position, once or continuously (for a mobile setup). Uses the standard `gpsd` service.

Laptop or desktop instead of a Pi

- macOS, Windows or Linux with **Python 3.10 or newer**.

- The `pygame-ce` package (`pip3 install pygame-ce`), the community edition of pygame. Classic `pygame` also works where it's available for your Python version.
- Any current web browser (Chrome, Safari, Firefox, Edge).

Network access used

SkyScope serves its pages on **TCP port 7700** on your local network. It connects out (HTTPS) to `opendata.adsb.fi` for aircraft, `api.planespotters.net` for photos, and `raw.githubusercontent.com` for airline logos.

3. Installing on a Raspberry Pi

1. **Prepare the SD card.** On another computer, install Raspberry Pi Imager. Choose your Pi model, **Raspberry Pi OS (64-bit)** (the version *with* desktop), and your SD card. When asked to customise settings, set:
   - **Hostname:** `skyscope` (so the settings page is at `http://skyscope.local:7700`)
   - **Username and password** of your choice
   - **Wi-Fi** network and password (skip if using Ethernet), and your time zone

   Write the card, put it in the Pi, connect the display, keyboard and mouse, and power on.

2. **Open a terminal** on the Pi (the black icon in the top bar) and bring the system up to date:

```
sudo apt update
sudo apt full-upgrade -y
```

3. **Install what SkyScope needs:**

```
sudo apt install -y python3-pygame git
```

4. **Download SkyScope** into your home folder:

```
git clone https://github.com/projectmoonpi/skyscope.git ~/skyscope
```

5. **Start it:**

```
python3 ~/skyscope/skyscope.py
```

   The radar fills the screen. Press `Q` to quit. The first start creates `~/skyscope/settings.json`, which holds your settings.

6. **Start it automatically when the Pi boots.** This copies the included launcher into your autostart folder, pointing it at your home folder:

```
mkdir -p ~/.config/autostart
sed "s#/home/pi#$HOME#" ~/skyscope/skyscope.desktop > ~/.config/autostart/skysc
```

Restart the Pi (`sudo reboot`); SkyScope should open full screen after the desktop appears.

> If it doesn't start on its own, add it to the desktop's own autostart list instead:
>
> ```
> mkdir -p ~/.config/labwc
> echo "python3 $HOME/skyscope/skyscope.py &" >> ~/.config/labwc/autostart
> ```

7. **Set it up from another device.** On a phone or computer on the same network, open `http://skyscope.local:7700/radar_config` (or `http://<pi-address>:7700/radar_config` ; the address is shown at the bottom of the Pi's screen) and continue with first-time setup.

Updating

```
cd ~/skyscope && git pull
```

Then restart SkyScope (or the Pi). Your `settings.json` is kept.

If the Pi struggles

Very long ranges (150 miles and beyond) near busy areas can mean 500+ aircraft. If the Pi display becomes jerky: use filters to show fewer aircraft, choose a shorter range, set the Pi's screen to 1280×720 (Screen Configuration in the Pi menu), or lower `fps` in `settings.json` . Another option is to show the web radar full screen in the Pi's browser instead of the Pi display:

```
python3 ~/skyscope/skyscope.py --web-only &
chromium --kiosk http://localhost:7700/radar
```

4. Installing on a Mac or Windows PC

SkyScope runs on a Mac or a Windows 10/11 PC as well as a Pi. On a computer you use the radar in your browser. An installer sets it up as a background program (a service) with icons to start and stop it, the same way as the US Train Tracker.

Step 1: Python (the installer sees to it)

SkyScope runs on Python 3.10 or newer. If the computer doesn't have it, SkyScope's installer offers to download it from python.org and install it: click **Install Python** (Mac; it asks for your Mac password) or **Yes** (Windows; installed for you only, no password). It checks the download is signed by the Python Software Foundation first.

To install Python yourself instead: python.org/downloads, then on a Mac open the downloaded `.pkg` and click through it; on Windows tick **Add python.exe to PATH** on the installer's first screen, then click **Install Now**.

Step 2: Download SkyScope

From projectmoonpi.com/downloads (or GitHub's latest release), download one file: **SkyScope-*version*-for-Mac.dmg** (a disk image) or **Install-SkyScope-*version*-for-Windows.exe** (the installer), for example SkyScope-1.9.0-for-Mac.dmg. (The .zip files there are for SkyScope 1.8.5's Update button.) On a Mac, the disk image also has a **READ ME FIRST** note and this guide.

Step 3: Run the installer

1. **Mac:** double-click the downloaded disk image, then **Install SkyScope** (the radar icon) in its window. It opens the installer's window, like the M.O.S.S. SD Card Creator's: it checks the Mac, asks its questions, then shows each step of the install. If the Mac hasn't got Python yet, it first offers to install it from python.org (click **Install Python**; macOS asks for your password). If macOS says it can't check the developer, open **System Settings → Privacy & Security**, scroll down to the message about it, and click **Open Anyway**. (On macOS 14 or older you can instead Control-click the file and choose **Open**.)
   Windows: double-click the downloaded **Install-SkyScope-*version*-for-Windows** (the radar icon). If Windows says it protected your PC, click **More info**, then **Run anyway**. It opens the installer's window, with the Project Moon Pi logo.

2. Answer its questions (tick boxes on one page, then **Install**):
    - **Where?** (Windows) **Install in** suggests a SkyScope folder of your own; type another or click **Browse...**. It can't go in Program Files (SkyScope installs just for you, without admin permission). A folder that already holds other things gets a SkyScope folder inside it.
    - **Phones too, or this computer only?** Choose phones if you'll open SkyScope on a phone or tablet, or use a SkyScope Satellite display. On Windows, a note then explains Windows' "allow this app to make changes" question for Install SkyScope.exe: click **Yes**. That opens the firewall for them, on home (private) networks only.
    - **Start when you log in? Yes** keeps SkyScope running whenever you're logged in; **No** starts it only when you start it (Start SkyScope on a Mac; Open SkyScope or Start SkyScope Service on Windows).
    - **Start and Stop on the Desktop?** They're in Applications → SkyScope (Mac) or the Start menu (Windows) either way.
3. It installs SkyScope (about a minute), showing each step, and starts it last. Click **Finish**: with **Open the radar now** ticked, the radar opens. If the Mac asks whether SkyScope (or Python) may accept incoming network connections, click **Allow**. You can delete the download afterwards (on a Mac, eject the disk image first).

Using it

- **Mac: Start SkyScope** (Applications → SkyScope, and the Desktop if you chose) starts it if it isn't running and opens the radar. **Stop SkyScope** shuts it down; the radar stops until you start it again.
- **Windows** (the Start menu, and the Desktop if you chose): **Open SkyScope** opens the radar, starting SkyScope first if it isn't running. **Start SkyScope Service** and **Stop SkyScope Service** only start and stop it in the background, without opening the radar.
- **Is it running?** Look for SkyScope's radar icon. On a Mac it's in the **menu bar** at the top of the screen, black or white like the icons beside it, and faint if SkyScope stops answering. It's there while SkyScope runs and goes away when SkyScope stops. Click it for a menu: the version, when it started and how many aircraft are in range, then **Open the radar**, **Settings** and **Stop SkyScope**. On Windows it's **by the clock** (new icons start under the **^** there; drag it onto the taskbar to keep it in sight), and it stays there once SkyScope has started, until you sign out: **green** while SkyScope runs, **red** when it's stopped. Click it for the radar (starting SkyScope first if it's stopped); right-click it for **Open the radar**, **Start SkyScope Service** and **Stop SkyScope Service**. A notice says when it has started or stopped.
- Activity Monitor (Mac) and Task Manager (Windows) list it as **SkyScope**. On a Mac, Start SkyScope says so if it's already running, and Stop SkyScope if it wasn't.

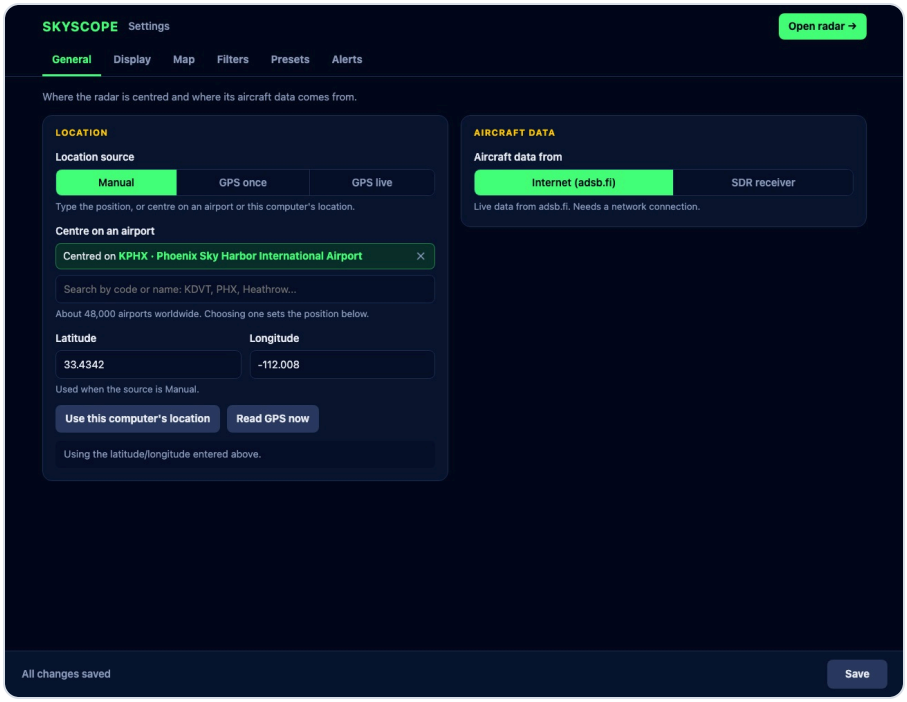
- **To update**, download a newer release and run its installer. When SkyScope is already installed, the installer asks what to do, with a button for each on a page of their own:
    - **Update** (from an older version): compares the installed files with the new ones and copies only those that differ. Your settings, FlightAware key and logo cache are kept.
    - **Repair** (the same version): checks every installed file, SkyScope's own and its Python parts, and puts back any that are damaged or missing, then sets up its apps or shortcuts again. Your settings, FlightAware key and logo cache are kept.
    - **Reinstall from scratch**: removes everything SkyScope put on the computer, its settings and logo cache too, then installs it new with its own settings. Only your FlightAware API key is kept. Use it when something's wrong that Repair doesn't fix.
- **Uninstall SkyScope** removes it and its settings; Python stays. If you've added a FlightAware API key, it asks whether to delete that too: untick the box to keep it, and installing SkyScope again puts it back. On a Mac it's in Applications → SkyScope. On Windows, open **Settings → Apps → Installed apps**, find SkyScope and click **Uninstall**, as for any program (**Uninstall SkyScope** is also in the folder SkyScope is installed in). On a Mac, the first time, macOS asks whether **Uninstall SkyScope** may access your Desktop folder: click **Allow** so it can remove its Desktop icons too.
- If it won't start, a message says so. Its messages are in `~/Library/Logs/SkyScope.log` on a Mac, or `skyscope.log` in the folder it's installed in on Windows (normally `%LOCALAPPDATA%\Programs\Project Moon Pi\SkyScope`) on Windows (in the folder you chose, if you chose another).

> **Where it goes:** on a Mac in `~/Library/Application Support/SkyScope`, on Windows in `%LOCALAPPDATA%\Programs\Project Moon Pi\SkyScope` (or a folder you choose), with its own copy of Python's packages (pygame-ce, the community edition of pygame, which has ready-made packages for the newest Python versions; pystray and Pillow for the icon).

> **Running from the source instead** (for developers): in a SkyScope folder from GitHub, run `python3 skyscope.py --web-only` (on Windows `py skyscope.py --web-only`) after `python3 -m pip install pygame-ce`, and stop it with `Ctrl`+`C`. `sh tools/make_mac_app.sh` or `tools\windows\create-shortcut.bat` makes desktop icons for that copy. `python3 skyscope.py --window 1280x720` shows the Pi-style display in a window.

5. First-time setup

Open the settings page (`http://<address>:7700/radar_config`). The tabs across the top hold every option; nothing changes until you press **Save** in the bar at the bottom, which shows "Unsaved changes" whenever there's something to save.



The General tab: where the radar is centred, and where its aircraft data comes from.

1. **Choose the centre.** On the **General** tab, with **Manual** selected, either search for an airport (by code or name, such as `KDVT`, `PHX` or `Heathrow`) and pick it, or type a latitude and longitude. On a laptop, **Use this computer's location** fills them in for you. A computer without Wi-Fi (a desktop on a network cable, or a virtual machine) can only guess from its internet address, often the middle of your provider's city: SkyScope then says it's a rough guess and waits for you to check the position and click **Save**.
2. **Pick your units** on the **Display** tab: distance (km, statute miles or nautical miles), speed and altitude.
3. **Pick a range** on the Display tab: the distance from the centre to the outer ring.
4. Press **Save**, then **Open radar** (top right) to see the result.

6. Using the radar

The scope

- **Range rings** are labelled with their distances along the east line. The range is the distance to the outer ring.
- **Aircraft** are drawn with a shape for their kind and a line showing where they'll be in about a minute. Labels give callsign, type and altitude; in busy areas some labels are left off to keep things readable (every aircraft is still in the list).
- The **yellow circle** marks the nearest aircraft. **Gold rings** mark watched aircraft and a **flashing red** symbol marks an emergency (see section 10).
- The green **sweep** is decorative; its speed and trail can be changed or switched off. On the settings page's Display tab, **Aircraft fade between sweeps** makes each aircraft brightest as the sweep passes, then dimming (never out) until it comes round again. **Fade rate** (1–100) sets how far they dim: 1 barely, 100 to a fifth. The fade takes one whole turn, so a slow sweep fades slowly; emergencies and watched aircraft never fade.
- **Radar colours** (Display tab): **Standard**, or **Green**, which draws the aircraft, their labels, runways, airspace and SIDs/STARs all in green, like an older ATC radar screen (emergencies and the watchlist keep their colours).
- Optional **dots on the rim** show aircraft just outside the range, in their direction.

On a phone or tablet

Click **Phone** at the bottom of the radar (or open the settings page's **About** tab): it shows a QR code and the radar's address on your network, such as `http://192.168.1.20:7700/radar`. Point the phone's camera at the code, or type the address into its browser. The phone must be on the same Wi-Fi. (On a phone or tablet itself, the radar leaves out the Phone and Fullscreen buttons.) If something stops phones getting through, a yellow note under the code says what to fix (see Troubleshooting).

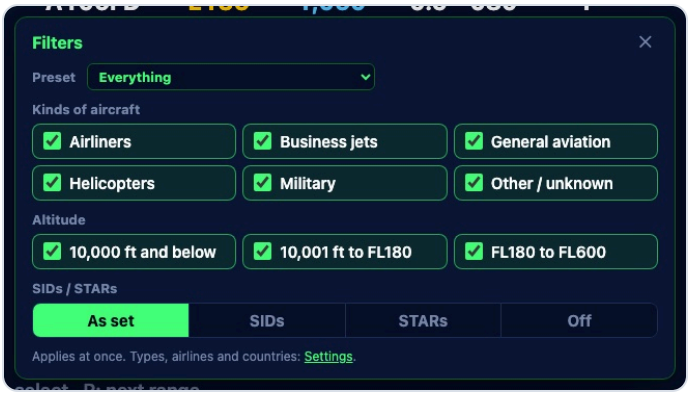
The panel

The top shows the centre, the number of aircraft, the data source and how fresh the data is. The **card** below shows the nearest aircraft, or the one you've selected: callsign, aircraft type and registration, the route of an airline flight (for example *KPHX → KOAK · Phoenix → Oakland*), altitude and climb or descent, distance and bearing, and ground speed. When Planespotters.net has a photo of that exact aircraft, a thumbnail appears; click it to open the full photo. Otherwise the box

shows a plane outline and *No image available* (or *Finding photo…* for a moment while it looks). The **list** shows every aircraft in range, nearest first, with units under each column heading.



A selected aircraft, with its airline, type, registration and a photo of that exact aircraft.



The web radar's Filters panel: kinds of aircraft, altitude blocks, presets and SIDs/STARs.

On the web radar

- **Click an aircraft** (its symbol or label) or a **row in the list** to select it. Brackets mark it on the scope and the card shows its details. Click it again, or press `Esc`, to clear the selection. Hovering over a row highlights that aircraft on the scope.
- ⚙ **Filters** at the bottom of the panel opens a panel to choose kinds of aircraft, altitude blocks, a preset and the SIDs/STARs switch. Changes apply at once.
- `R` or `Space`: next range. `F`: full screen. Clicking empty space never changes the range.
- **Settings** and **Guide** links at the bottom open the settings page and this guide.

On the Pi display

With a keyboard attached: `R`/`Space`/`→` next range, `←` previous range, `U` distance units, `W` runways, `S` sweep, `A` symbol style, `L` logos, `D` rim dots, `G` aircraft on the ground, `F` full screen, `Q` quit. Everything else is on the settings page.

7. The settings page

Tab	What's on it
General	Where the radar is centred (manual, an airport, or GPS), where aircraft data comes from (internet or your own receiver), flight routes (optionally with a FlightAware key), and whether other devices may change settings (see below).
Display	Range; aircraft symbols (silhouettes or simple triangles); colouring aircraft by kind; airline logos; the aircraft photo; rim dots; aircraft on the ground; radar colours (standard or green); units; the sweep, and whether aircraft fade between sweeps (and how much).
Map	Airports and runways, extended centrelines, airspace, and SIDs/STARs per airport. See section 8.
Filters	Which aircraft appear. See section 9.
Presets	Named sets of filters: apply, save and delete.
Alerts	Emergency highlighting and the watchlist. See section 10.
About	Updates (is there a newer SkyScope?), a QR code for phones and tablets, SkyScope's version, the addresses of this radar (including the one to give a Satellite display), links to this guide and the project pages, data sources and credits.

The page works on phones as well as computers. Anyone on your network can open it and look.

Changes from other devices

The computer running SkyScope can always change everything. Phones, tablets and other computers on your network can always look at the radar and the settings; whether they can also change settings, filters and the range is up to **Changes from other devices** on the General tab:

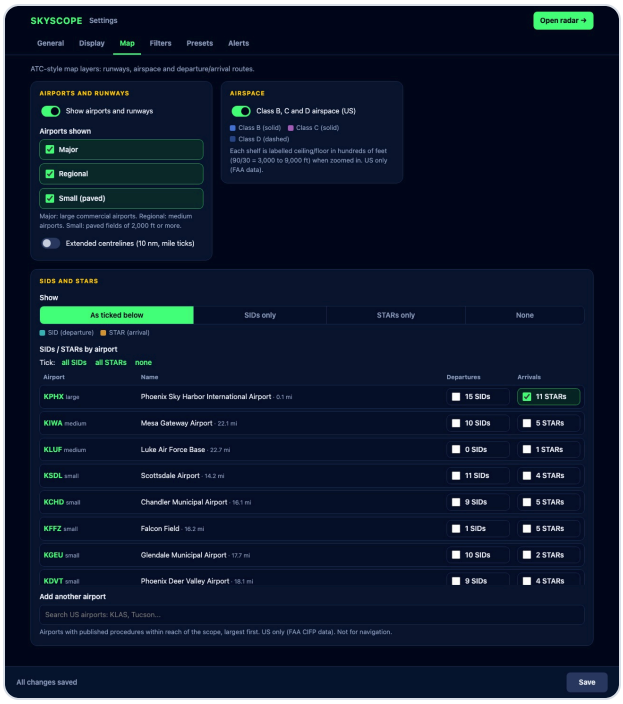
- **View only** (the starting choice): they can look at the radar, but nothing they do changes SkyScope. Their radar has no **Settings** button (the settings address takes them back to the radar), and its Filters panel is greyed out.
- **With a PIN**: set a PIN of 4 to 8 digits. On another device, **Settings** asks for the PIN first: the right PIN opens the settings, and **Cancel** goes back to the radar with nothing changed. A device remembers the PIN for a year.
- **Allowed**: any device on your network can make changes.

Only the computer running SkyScope can change this choice or the PIN.

Updates

SkyScope asks GitHub a few times a day whether there's a newer version (only while **Check for a newer SkyScope** is on, in About; the request says only which version is asking). When there is, a note appears on the radar, the Mac menu bar icon's menu offers it, and the **Updates** card in About has an **Update to** button. **Check now** asks at once. **Update to** (on the computer running SkyScope) downloads the new version in the background, checks it is the file GitHub published, unpacks it and opens its installer in front: click **Update** there, and SkyScope is updated and started again with your settings kept. **Or download it yourself** gets the installer to run by hand. On a Raspberry Pi, the card links to the release page.

8. The ATC map



The Map tab: airport types, centrelines, airspace, and SIDs/STARs chosen per airport.

Airports and runways

Runways are drawn to scale. Runways of 5,000 ft or more get **extended centrelines** reaching 10 nautical miles out, with a tick every mile and a longer tick every five, as on an approach-control display. Runway numbers appear at the ends when zoomed in. Choose which airports appear: **Major**, **Regional** and **Small (paved)**, each on or off. Airport data covers the whole world.

Airspace (US)

Class B (solid blue), Class C (solid magenta) and Class D (dashed blue) outlines from the FAA. When zoomed in, each shelf is labelled with its ceiling over its floor in hundreds of feet: `90/30` means from 3,000 to 9,000 feet.

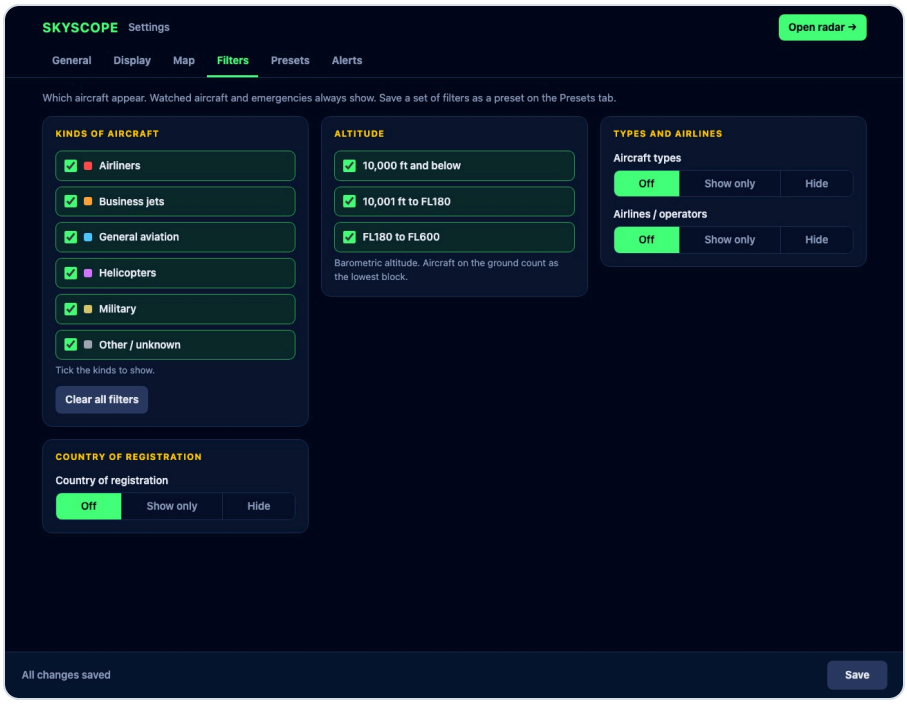
SIDs and STARs (US)

The Map tab lists every airport with published departure (SID) and arrival (STAR) routes within reach of the scope, larger airports first, with how many each has. Tick **SIDs**, **STARs** or both for each airport you want; use **Add another airport** for one further away. Departures are drawn in teal

and arrivals in amber, with the waypoints (△) named. The **Show** switch (also in the web radar's Filters panel) hides them all, or shows only SIDs or only STARs, without losing your choices.

> Parts of some routes are flown on a heading with no fixed track ("climb on heading 078, expect vectors"). Those parts can't be drawn, so a route may start or end away from the runway.

9. Filters and presets



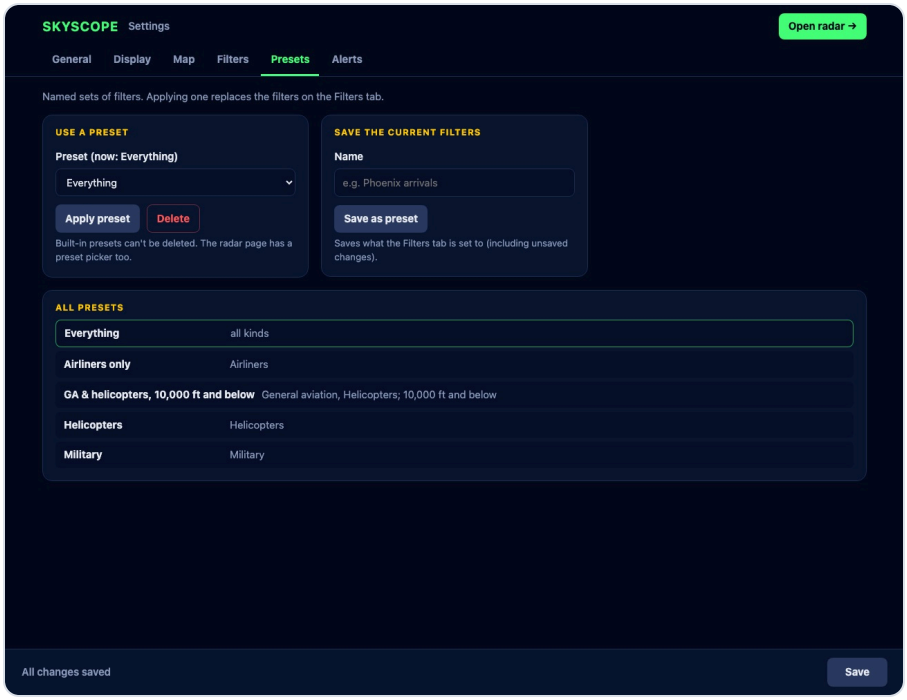
The Filters tab. Tick what you want to see.

- **Kinds of aircraft:** airliners, business jets, general aviation, helicopters, military, and other/unknown. A light aircraft flying a flight school's callsign counts as general aviation.
- **Altitude:** 10,000 ft above the ground and below; from there to FL180; FL180 to FL600. The lowest block counts from the ground at the radar's centre, because helicopters and light aircraft fly over high country well above 10,000 ft MSL: in Denver (about 5,200 ft) it reaches 15,200 ft, at sea level 10,000 ft. The ground elevation is found automatically (from the GPS, or looked up for the centre) or entered by hand on the **General** tab, which shows the value in use.
- **Aircraft types:** show only, or hide, a list of type codes such as `B738, A32*, C172` (`*` matches anything).
- **Airlines:** show only, or hide, by the three-letter code at the start of the callsign, such as `AAL, SWA, UAL`.
- **Country of registration:** worked out from each aircraft's transponder address, so it works even when the registration is unknown.

While any filter is on, the panel says, for example, "35 of 134 aircraft (filtered)". Emergencies and watched aircraft are always shown.

Presets

Built-in presets: *Everything*, *Airliners only*, *GA & helicopters, 10,000 ft AGL and below*, *Helicopters* and *Military*. To make your own, set the filters you want, then on the **Presets** tab type a name and press **Save as preset**. Apply presets on the Presets tab or from the web radar's Filters panel.



The Presets tab lists every preset with a summary of what it shows.

10. Alerts and the watchlist

- **Emergencies:** aircraft squawking 7700 (emergency), 7600 (radio failure) or 7500 (hijack), or reporting an emergency status, flash red with a pulsing ring, are labelled first, move to the top of the list and get a red banner in the panel.
- **Watchlist** (Alerts tab): callsigns, registrations, transponder addresses or type codes, with `*` wildcards, for example `N123AB, ENY*, B52`. Matching aircraft get a gold ring and a ★ in the list, are labelled first, and optionally get a gold banner while in range.

11. Your own receiver and GPS

Receiving aircraft yourself

1. Plug an RTL-SDR dongle with a 1090 MHz antenna into the Pi. Place the antenna as high as possible with a clear view of the sky.

2. Install `readsb`, the decoding software, and tell it your location:

```
sudo bash -c "$(wget -O - https://github.com/wiedehopf/adsb-scripts/raw/master/
sudo readsb-set-location <your-latitude> <your-longitude>
```

3. On the settings page's **General** tab, set **Aircraft data from** to **SDR receiver** and save. With the location left at `auto`, SkyScope finds readsb (or dump1090/PiAware) by itself; the status line says which it's reading.

Range now depends on your antenna and its view of the sky. Aircraft types and names may be blank for some aircraft unless readsb's aircraft database is enabled.

Sharing with FlightAware (optional)

Your receiver can also feed FlightAware with its free **PiAware** software, installed alongside readsb by following FlightAware's instructions. SkyScope keeps reading readsb as before. In return, FlightAware gives feeders a free Enterprise account and doubles the free FlightAware lookup allowance to about $10 a month, which suits SkyScope's flight routes.

Flight routes

ADS-B carries no flight plan, so SkyScope finds routes elsewhere and shows them on the card (for example *KPHX → KOAK · Phoenix → Oakland*). Turn them on or off on the **General** tab, card **Flight routes**, below **Aircraft data**.

- **Without a key**, airline flights are looked up by flight number in two free community databases, adsbdb.com and hexdb.io. Flight numbers get reused and the databases go out of date, so a route shows only when it fits where the aircraft is. Expect routes on a minority of airline flights, and none for private or VFR traffic.
- **With a FlightAware AeroAPI key**, the card shows the actual filed route of the aircraft on it, private IFR flights included. A Personal key from flightaware.com/aeroapi includes about $5 of lookups a

month free ($10 if you feed FlightAware). Paste it into the key box and choose a daily limit. SkyScope looks up only the aircraft on the card, once per flight, and stops at the limit. The card's status line shows how many lookups are used today. Check FlightAware's fee table to choose a limit that stays inside your allowance.

The key is kept in `settings.json` on this computer. It is never shown on the settings page or sent to the radar page.

Centring on a GPS position

1. Plug in a USB GPS receiver and install the GPS service:

   ```
   sudo apt install -y gpsd
   ```

   USB receivers are usually detected automatically. For a GPS add-on board on the serial port, set `DEVICES="/dev/serial0"` in `/etc/default/gpsd` and run `sudo systemctl restart gpsd`.

2. On the **General** tab, choose **GPS once** (reads the position when SkyScope starts: for a fixed installation) or **GPS live** (keeps re-centring every few seconds as you move), and save. The status line shows the fix and number of satellites.

12. Troubleshooting

Problem	What to try
The panel says "No data" or "Waiting for data"	Check the Pi's internet connection. With an SDR receiver, read the status line on the General tab: it says whether readsb was found and whether its data is current.
"HTTP Error 429: Too Many Requests"	adsb.fi allows one request per second. Make sure only one copy of SkyScope runs per network address; it recovers by itself.
A phone's browser times out on the radar's address (Windows)	Windows' firewall is keeping it out. Look at the yellow note under the QR code (Phone button, or About): usually Windows calls your network **Public**; make it **Private** in Settings › Network & internet › Wi-Fi (or Ethernet) › your network › Network profile type. If the note says the firewall opening is missing, run Install SkyScope again, choose **Repair** with "Let phones..." ticked, and click **Yes** when Windows asks. Some antivirus programs have a firewall of their own that needs SkyScope allowed too.
The settings page won't open from another device	Use the address shown at the bottom of the Pi's screen. Both devices must be on the same network. `.local` names can be slow on some devices; use the numeric address instead.
An aircraft is in the list but not on the scope	Aircraft are drawn once they're inside the outer ring; the list shows the same set. Turn on rim dots (Display tab) to see aircraft just beyond the range.
No route for a flight	Without a FlightAware key, routes come from community databases by flight number, which don't know every flight; SkyScope shows one only when it fits where the aircraft is. A FlightAware key gives the filed route of the card's aircraft. See Flight routes.
“No image available” for an aircraft	Photos only appear when Planespotters.net has one of that exact aircraft; many light aircraft and helicopters have none. To hide the photo box altogether, turn off the aircraft photo on the settings page's Display tab.

No SIDs or STARs appear	Tick SIDs or STARs for at least one airport on the Map tab, and check the Show switch isn't set to None. Only US airports have this data.
"Use this computer's location" puts you in the middle of town	That computer has no Wi-Fi to find itself by (a desktop on a cable, or a virtual machine), so it guessed from the internet address. Pick your airport or type the position instead.
"Use this computer's location" is missing	Browsers only share location with the computer running SkyScope (`localhost`). On a Pi, type the position, pick an airport, or use GPS.
The Pi display is jerky	See If the Pi struggles in section 3.
Still stuck, or something else	See the common questions at projectmoonpi.com/support, or email support@projectmoonpi.com: say which computer SkyScope runs on (Mac, Windows or Raspberry Pi), its version (settings page › About) and what you saw. Its log helps too (see section 4).

13. Quick reference

Address	What it is
`http://<address>:7700/radar`	The web radar (also the bare `:7700` address)
`http://<address>:7700/radar_config`	The settings page
`http://<address>:7700/radar_config#about`	The settings page's About tab
`http://<address>:7700/guide`	This guide
`http://<address>:7700/api/satellite`	The feed SkyScope Satellite displays read

Command	What it does
`python3 skyscope.py`	Pi display, full screen. It opens with the Project Moon Pi start-up screen (the badge's light show, the version and the credit, about four seconds) while the first aircraft load; any key skips it. The browser radar shows the same start-up screen once per tab
`python3 skyscope.py --window 1280x720`	Pi display in a window
`python3 skyscope.py --web-only`	Web radar only, and opens it in the browser
`python3 skyscope.py --no-splash`	Starts without the start-up screen

Ranges	Choices (distance to the outer ring)
Statute miles	2, 5, 10, 15, 25, 50, 75, 100, 150, 200, 250
Kilometres	5, 10, 15, 25, 50, 100, 150, 200, 300, 400
Nautical miles	2, 5, 10, 20, 40, 60, 80, 100, 150, 200

All settings are stored in `settings.json` in the SkyScope folder. The README lists every entry for anyone who prefers to edit it by hand.

14. SkyScope Satellite and Mini

Two smaller companions run on an inexpensive 4.3-inch touch display with a built-in ESP32-S3 microcontroller (the Sunton ESP32-8048S043, 800 × 480). Each has its own project page with installation instructions.

	SkyScope Satellite	SkyScope Mini
What it is	A second screen for this SkyScope: it shows the same radar, range, filters and map	A small radar of its own; no Raspberry Pi needed
Data	From this SkyScope over your network	From adsb.fi over Wi-Fi
Map	Everything this SkyScope shows: runways, airspace, SIDs and STARs	Runways at major airports
Set up from	This settings page (it follows SkyScope's settings)	Its own small web page
Project page	github.com/projectmoonpi/skyscope-satellite	github.com/projectmoonpi/skyscope-mini

Connecting a Satellite

A Satellite looks for `skyscope.local` on your network unless told otherwise. Either name the SkyScope computer `skyscope` (on a Raspberry Pi: `sudo raspi-config` → System Options → Hostname), or type the address shown under **Satellite address** on the **About** tab into the Satellite's setup page. SkyScope must be running with the settings page on (port 7700, the default). Several Satellites can follow one SkyScope.

15. Credits and data sources

SkyScope grew out of ESP32-Plane-Radar by MatixYo (MIT licence), a radar for a small round display on an ESP32 microcontroller.

Data	Source	Terms
Live aircraft	adsb.fi open data, or your own receiver	adsb.fi's terms (one request per second)
Airports and runways	OurAirports	Public domain
US airspace	FAA Class Airspace data	Public domain
US SIDs and STARs	FAA CIFP (updated every 28 days)	Public domain
Country from transponder address	ICAO address allocations	Facts
Airline logos	Jxck-S/airline-logos, downloaded when first needed	Airlines' trademarks
Aircraft photos	Planespotters.net, shown with the photographer's credit	Photographers' copyright
Flight routes	adsbdb.com and hexdb.io, by flight number	Free community data
Filed flight routes (optional)	FlightAware AeroAPI, with your own key	FlightAware's AeroAPI terms
Ground elevation	Open-Meteo elevation API (Copernicus DEM GLO-90)	CC BY 4.0

Software	Source	Licence
Language and display	Python, pygame (or pygame-ce)	PSF; LGPL 2.1
Mac menu bar icon (installed copies)	pystray, Pillow, and PyObjC on a Mac	LGPL 3; MIT-CMU; MIT
Secure connections (installed copies)	truststore by Seth Larson: web connections checked with the computer's own certificates	MIT
QR codes for phones	QR Code Generator for JavaScript by Kazuhiko Arase (qrcode-generator 1.4.4)	MIT
Windows installer	C# on Microsoft's .NET Framework 4.8, built with Mono's C# compiler	Part of Windows; MIT
Built with	Claude Code (Anthropic): designed, written and tested with Claude	

> **Not for navigation.** Aircraft positions may be delayed, incomplete or wrong, and map layers are simplified. Never use SkyScope for flight planning or navigation.

SkyScope was designed and created by Project Moon Pi, built with Claude Code (Anthropic), and is free software under the MIT licence. Website: projectmoonpi.com · Help: support@projectmoonpi.com · Source: github.com/projectmoonpi/skyscope.